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Honours B.Sc. in **Artificial Intelligence** Oshawa, Ontario, Canada

Specialized in **Applied Machine Learning** & **Reliable AI Systems**

Objective

AI Research Assistant building reliable ML systems across computer vision, LLM safety, RAG, and cloud deployment. Seeking an AI/ML opportunity to contribute to safe, production-grade systems.

Education

Durham College, B.Sc. in Artificial Intelligence | **Honour Roll:** Fall 2025, Winter 2026 **Expected April 2027**
Relevant Courses: Machine Learning, Computer Vision, Natural Language Processing, Statistics, Linear Algebra, Systems Analysis and Design, Computer Architecture.

Experience

Research Assistant, Durham College AI Hub May 2026 – **Present**

- Built the agentic core for a **multi-persona LLM chatbot** on AWS Lambda and Bedrock, supporting persona routing, multi-turn context, and SSM-managed prompts; load-tested to **1.8s p95 latency** with Locust.
- Shipped a safety redline layer that detects **high-risk** language before LLM classification and triggers immediate human escalation, with deterministic matching.
- Implemented retrieval-augmented generation (**RAG**) with Amazon Bedrock Knowledge Bases and Amazon Titan Text Embeddings V2, configuring document ingestion, chunking, and retrieval to ground chatbot responses in domain-specific source material.
- Co-authored and presented a **technical advisory report** with a three-person team to the client, translating testing findings and technical risks into actionable implementation recommendations.

Projects

Financial News Agent | Python, FastAPI, Claude API, Supabase, SvelteKit **Repository**

- Built an **agentic financial assistant** with Claude API and autonomous tool use, allowing the model to call price, news, financial, and insider-trade tools based on user intent.
- Secured **FastAPI** endpoints with Supabase **JWT authentication**, row-level security, per-IP rate limiting, and input sanitization.
- Built a **SvelteKit** frontend with SSE streaming, real-time tool-call progress, and portfolio views.

Vision Transformer Research | Python, PyTorch, Scikit-learn, Pandas, OpenCV, Matplotlib, Seaborn **Repository**

- Benchmarked **8+ vision architectures**, including DINOv2, ViT, SwinV2, ResNet, and YOLO, for binary skin cancer classification using stratified 5-fold cross-validation.
- Achieved **~92% sensitivity** at a clinically relevant threshold using ablation studies, freezing strategies, and ensemble evaluation.
- Built preprocessing, augmentation, and training pipelines for a quality-controlled 906-case skin-lesion dataset, enabling **consistent** and **reproducible** evaluation across models.

Temperature Modelling | Python, Pandas, Scikit-learn, XGBoost, FastAPI, Streamlit, GCP, Docker **Repository**

- Developed an **XGBoost** regression model for temperature prediction using historical weather data, applying cross-validation and regularization to reduce overfitting and improve generalization.
- Productionized the model as a **Dockerized FastAPI** service on **Google Cloud Run** and connected it to a Streamlit interface for on-demand predictions.

Technologies

Languages: Python, C++, SQL, TypeScript.
Frameworks: PyTorch, JAX, TensorFlow, Scikit-learn, XGBoost, OpenCV, LangChain, LangGraph, Anthropic SDK.
Data: Pandas, NumPy, Matplotlib, Seaborn, Plotly.
Tools: Git, Pytest, Locust, Ragas, Tableau, Power BI.
Cloud & Infra: AWS (EC2, S3, Lambda, Bedrock, API Gateway), GCP, Docker, Linux.